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¹ The Act of 20th July 2018 on the Law on Higher Education and Science:

Art. 312, section 3. Should a student be suspected of committing an act referred to in Art. 287 section 2 items 1–5, the rector shall forthwith order an enquiry.

Art. 312, section 4. If the evidence collected during an enquiry confirms that the act referred to in section 5 has been committed, the rector shall suspend the procedure for the awarding of a professional title pending a judgement of the disciplinary committee and submit formal notification on suspicion of committing a criminal offence.



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ABSTRACT

This document is a $\text{\LaTeX}$ template for theses prepared at the Faculty of Electronics, Telecommunications and Informatics, Gdańsk University of Technology. The structure of the thesis should comply with the guidelines specified in [Rector's Ordinance No. 45/2024](#) of 15 November 2024. These guidelines define the general rules for preparing theses and the editorial requirements. This template reflects, clarifies, and, in selected issues, extends these rules (including the numbering of tables, figures, equations, and citations). It also facilitates work with cross-references, captions, and the table of contents. It provides a preconfigured set of $\text{\LaTeX}$ packages and practical guidance for the author, i.e., what to place in individual chapters, how to describe methods and results, and how to prepare figures and tables.

The main project file is `main.tex`. The formatting settings are collected in a separate class file `_formatting.cls`, which contains all the required elements that build the thesis's structure. Additional formatting hints are described in separate chapters of the template. The document structure comprises four main chapters stored in the *chapters* directory and an appendix with an example Python code. Each chapter is placed in a separate `.tex` file. The bibliography is managed with the `biblatex (biber)` package, in accordance with the ISO 690 standard, ensuring square-bracketed numbering in citation order. The title page and the student's declaration are attached as PDF files generated in the MyGUT (Moja PG) portal and inserted into the main file using the `\includepdf` command.

The entire project was optimized for compilation in the Overleaf environment with the necessary use of the LuaLaTeX engine (TeX Live distribution, version at least 2022). The default font used in the form is Arial in size 10 pt, and its files are included in the `fonts` directory, which allows the project to be compiled in any environment. This approach combines automation of formatting with the flexibility of $\text{\LaTeX}$, ensuring compatibility with the standards of Gdańsk University of Technology. Please, remember that the document must meet the requirements of anti-plagiarism regulations. The generated PDF file cannot be graphic or encrypted in terms of text; it can also not use non-standard font encoding (e.g., by replacing character codes) or 'combining' diacritics from several characters into one glyph. Summary: this template aims to minimize the author's technical workload so they can focus on the content, while maintaining a uniform editorial style for all theses at the ETI Faculty. Any comments regarding the template should be addressed to its author, Ph.D. Eng. Sebastian Dziedziewicz, e-mail: sebastian.dziedziewicz@pg.edu.pl. Below are example keywords that may be used in a thesis.

Keywords :

machine learning, PID control, embedded systems, analog circuits, modulation, distributed systems

Field of science and technology in accordance with OECD requirements:

engineering and technology; electrical, electronic and information engineering; robotics and automation; telecommunications.

STRESZCZENIE

Niniejszy dokument jest szablonem $\text{\LaTeX}$ dla prac dyplomowych realizowanych na Wydziale Elektroniki, Telekomunikacji i Informatyki Politechniki Gdańskiej. Struktura pracy powinna być zgodna z wytycznymi określonymi w [Zarządzeniu Rektora nr 45/2024](#) z 15 listopada 2024 r. Wytyczne te definiują ogólne zasady przygotowania prac dyplomowych oraz wymagania edytorskie. Niniejsza formatka odzwierciedla, precyzuje i w wybranych miejscach rozszerza te zasady (m.in. dotyczące numerowania tabel, rysunków, wzorów i cytowań). Ułatwia również pracę z odwołaniami, podpisami i spisem treści. Zawiera wstępnie skonfigurowany zestaw pakietów $\text{\LaTeX}$ oraz praktyczne wytyczne „dla autora”, tj. co umieszczać w poszczególnych rozdziałach pracy, jak opisywać metody i wyniki oraz w jaki sposób przygotowywać rysunki i tabele.

Głównym plikiem projektu jest `main.tex`. Ustawienia formatowania zebrano w oddzielnym pliku klasy `_formatting.cls`, który zawiera wszystkie wymagane elementy budujące strukturę pracy. Dodatkowe wskazówki dotyczące formatowania pracy dyplomowej opisano w osobnych rozdziałach formatki. Struktura dokumentu obejmuje cztery główne rozdziały umieszczone w katalogu *chapters* oraz dodatek z przykładowym kodem w języku Python. Każdy rozdział znajduje się w osobnym pliku `.tex`. Bibliografia jest zarządzana przy użyciu pakietu `biblatex` (`biber`) zgodnie z wytycznymi normy PN-ISO 690:2012, co zapewnia numerację w nawiasach kwadratowych zgodnie z kolejnością cytowań. Stronę tytułową oraz oświadczenie studenta dołącza się jako pliki PDF wygenerowane w systemie MojaPG i wstawia do dokumentu poleceniem `\includepdf` w pliku głównym.

Całość projektu zoptymalizowano pod kompilację w środowisku Overleaf z koniecznym użyciem silnika LuaLaTeX (dystrybucja TeX Live w wersji co najmniej 2022). Domyślną czcionką zastosowaną w formatce jest Arial w rozmiarze 10 pt, a jej pliki zostały dołączone w katalogu `fonts`, co umożliwia kompilację projektu w dowolnym środowisku. Takie rozwiązanie łączy automatyzację formatowania z elastycznością $\text{\LaTeX}$ oraz zapewnia pełną zgodność ze standardami Politechniki Gdańskiej. Należy pamiętać, że dokument musi spełniać wymogi przepisów antyplagiatowych — wygenerowany plik PDF nie może być w zakresie tekstu plikiem graficznym ani szyfrowanym; nie może także wykorzystywać niestandardowego kodowania czcionek (np. poprzez podmianę kodów znaków) ani „składania” znaków diakrytycznych z kilku znaków w jeden glif. Podsumowując, celem szablonu jest zminimalizowanie pracy technicznej autora, tak aby mógł on skupić się na treści, przy jednoczesnym zachowaniu jednolitego stylu edytorskiego wszystkich prac powstających na Wydziale ETI. Wszelkie uwagi dotyczące szablonu prosimy kierować do autora formatki: dr. inż. Sebastiana Dziedziewicza, e-mail: sebastian.dziedziewicz@pg.edu.pl. Poniżej zamieszczono przykładowe słowa kluczowe, które mogą zostać wykorzystane w pracy.

Słowa kluczowe:

uczenie maszynowe, sterowanie PID, systemy wbudowane, układy analogowe, modulacja, systemy rozproszone

Dziedzina nauki i techniki, zgodnie z wymogami OECD:

nauki inżynierskie i techniczne; elektrotechnika, elektronika, inżynieria informatyczna; robotyka i automatyka; telekomunikacja.

LIST OF IMPORTANT SYMBOLS AND ABBREVIATIONS

Basic Physical Quantities and Mathematical Expressions

j	$[\cdot]$	– imaginary unit
∇	$[\cdot]$	– nabla operator
$\vec{i}_x, \vec{i}_z, \vec{i}_y$	$[\cdot]$	– unit vectors of the Cartesian coordinate system
$\vec{F}$	$[\text{N}]$	– force
t	$[\text{s}]$	– time
m	$[\text{kg}]$	– mass
R	$[\Omega]$	– resistance
$\mathbf{S}$	$[\cdot]$	– scattering matrix

Physical and mathematical constants

$\pi \approx 3.14159$	$[\cdot]$	– number π
$g \approx 9.80665$	$[\frac{\text{m}}{\text{s}^2}]$	– gravitational acceleration

Terminology used in the thesis

<code>\section{}</code>	– creates a chapter title
<code>\subsection{}</code>	– creates a subsection title
<code>\cite{}</code>	– reference to a bibliography item
<code>\ref{}</code>	– reference to a figure, table, or equation
<code>\eqref{}</code>	– reference to an equation (with round bracket)
<code>\begin{figure}</code>	
...	– inserting a graphic object
<code>end{figure}</code>	
<code>\begin{table}</code>	
...	– inserting a table
<code>end{table}</code>	
<code>\begin{equation}</code>	
...	– inserting a numbered equation
<code>end{equation}</code>	
<code>\textbf{}</code>	– boldface text
<code>\emph{}</code>	– italic emphasis
<code>\footnote{}</code>	– insert a footnote
<code>\includegraphics{}</code>	– insert an image
<code>\label{}</code>	– object label
<code>\item</code>	– list item

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1. INTRODUCTION AND BACKGROUND OF THE THESIS

The continuous development of technology ... This chapter should present the problem to be analysed or solved within the thesis, together with a justification for undertaking the topic. It should briefly discuss the current state of the art and existing solutions related to the selected issue.

It is recommended to include a theoretical description of the problem, definitions of basic concepts, and a review of existing solutions - highlighting their strengths and weaknesses. It is also important to clearly formulate the thesis topic in line with the approved diploma data in the MyGUT (Moja PG) portal. The entire text should be prepared with editorial consistency, linguistic and terminological correctness, and overall coherence.

1.1. *State of the art*

For many years, there has been a dynamic increase in the number of publications devoted to ... This section should provide a concise literature review covering 10–20 items. It should include both classic works forming the foundations of the field and the most recent publications from the last five years. The aim is to present, briefly yet coherently, the current state of knowledge and solutions related to the thesis topic, together with their initial analysis and assessment.

Citations in the text should be inserted with the command `\cite{key}`, e.g. [1]. Bibliography should be prepared in accordance with the guidelines of the **ISO 690** standard, using the file `_bibliography.bib` where all sources included in the thesis are defined. The references [2] will be collected in the Bibliography, in accordance with editorial requirements and good thesis-writing practice.

The numbering of bibliography entries is automatic and follows the order of appearance in the text [3]. If another numbering scheme is required, it can be adjusted via the `biber` backend, e.g., in the file `_formatting.cls`, although the default setting is recommended.

1.2. *Aim and Scope of the Thesis*

The subject of this thesis is ... The text should be stated clearly and concisely, including the **topic** and the precise aim of the thesis. If necessary, a thesis statement (hypothesis) should be formulated. Then, briefly present prior work in the field, the adopted technical assumptions, and provide a short preview of the content of the subsequent chapters.

In the introduction, it is worth outlining the current state of knowledge and technical solutions, identifying unexplored areas and key research and design challenges. It should also be justified that the subsequent parts of the thesis develop the topic in an original and well-founded manner.

In the case of multi-author theses, authorship of individual parts must be clearly assigned. One author should prepare each subsection, and the authors of chapters or subsections should be listed in the appropriate place in the document.

2. DESCRIPTION OF THE METHODOLOGY

This chapter should present a detailed description of the methods, tools, and techniques used in the thesis. It should develop the theoretical issues outlined in the introduction (see Chapter 1) and clearly indicate how the author approached achieving the thesis objective.

First, the applied research or design methods should be described, together with their justification and the context of use. In empirical studies, the study group, sampling procedure, measurement instruments, and research procedures should be specified. For design or engineering theses, the technologies, programming environments, libraries, and frameworks should be presented, along with a brief discussion of the technical assumptions adopted.

Next, the project stages should be presented in a logical order - from requirements analysis and initial concepts, through implementation, to testing and validation of the results. It is advisable to supplement the description with block diagrams, UML diagrams, mathematical formulas, figures, or tables to clarify the outcomes.

Example structure of the methodology:

1. Characteristics of methods and tools - a description of selected research techniques, algorithms, programming environments, or devices.
2. Description of the implementation stages - actions presented in sequence, together with the justification of their use.
 - a) Stage 1: requirements analysis.
 - b) Stage 2: solution design.
 - Stage 3: implementation and integration.
 - Stage 4: testing and verification of results.
3. Summary - a brief overview of the adopted methodology and its relevance to achieving the thesis objectives.

The entire content should be prepared coherently so that the reader can clearly follow the logic and sequence of actions leading to the thesis's results. Footnotes may be inserted using the command `\footnote{Text.}`, e.g. ¹

¹Their numbering follows the order of appearance in the thesis.

3. RESULTS

This chapter should present the results obtained during the preparation of the thesis. They should constitute a logical extension of the theoretical part and the design methodology. The description should include the obtained outcomes. It is also necessary to explain what they mean, i.e., provide their interpretation. The description should evaluate the results, indicating their significance and reliability. It is also advisable to compare them with other solutions, theoretical data, or expected outcomes. Depending on the kind of thesis, this chapter may also include a detailed list of the methods, tools, or technologies used, along with examples of their practical applications.

3.1. *Illustrative elements and results analysis*

This subsection presents example ways of displaying results, including:

- mathematical formulas (with numbering on the right-hand side of the column),
- figures (with captions placed under the illustration, continuous numbering),
- tables (with the title above the table and units in a separate column),
- as well as the rules for referring to these elements in the text.

Using the command `\noindent`, it is possible to continue a paragraph (without indentation).

3.1.1. *Editing formulas*

Mathematical formulas should be inserted using the equation environment, which automatically assigns numbering, e.g.:

$$\mathbf{A}x = b, \tag{3.1}$$

where:

- $\mathbf{A} \in \mathbb{R}^{n \times n}$ - coefficient matrix,
- $\mathbf{x} \in \mathbb{R}^n$ - vector of unknowns,
- $\mathbf{b} \in \mathbb{R}^n$ - right-hand-side vector.

Each formula should be referenced in the text using the command `\eqref{}`, as shown in equation (3.1). All symbols used in the formula should be unambiguously explained directly below it. Variables and physical quantities should be written in italics, e.g., x , whereas matrices and vectors should be in bold mathematical font, e.g., $\mathbf{A}$. One should also remember the rules for writing numerical values (no space before the minus sign), e.g., -20 , and separating units with a space, e.g., 5 V , or using the command `\:` inside a mathematical operator, e.g., $10\frac{\text{m}}{\text{s}^2}$. If possible, equations should not be inserted directly in the paragraph. If a system of equations needs to be presented, the cases environment may be used:

$$\begin{cases} x + y = 5, \\ 2x - y = 1. \end{cases} \tag{3.2}$$

3.1.2. *Plots and graphics*

Each plot or figure should have a number and a caption placed below the illustration. Please note here that the font size (Arial 9 pt), axis proportions, and overall aesthetics of the plot should be

preserved. size (Arial 9 pt), axis proportions and the overall aesthetics of the plot. Vector formats are recommended (e.g., .eps, .pdf, .svg), as they preserve full quality when scaling. Raster graphics (e.g., .png, .jpg) from low-resolution screenshots should be avoided. A figure should be inserted using the figure environment with the `[!htbp]` option to keep its position in a given place in the text, e.g.:

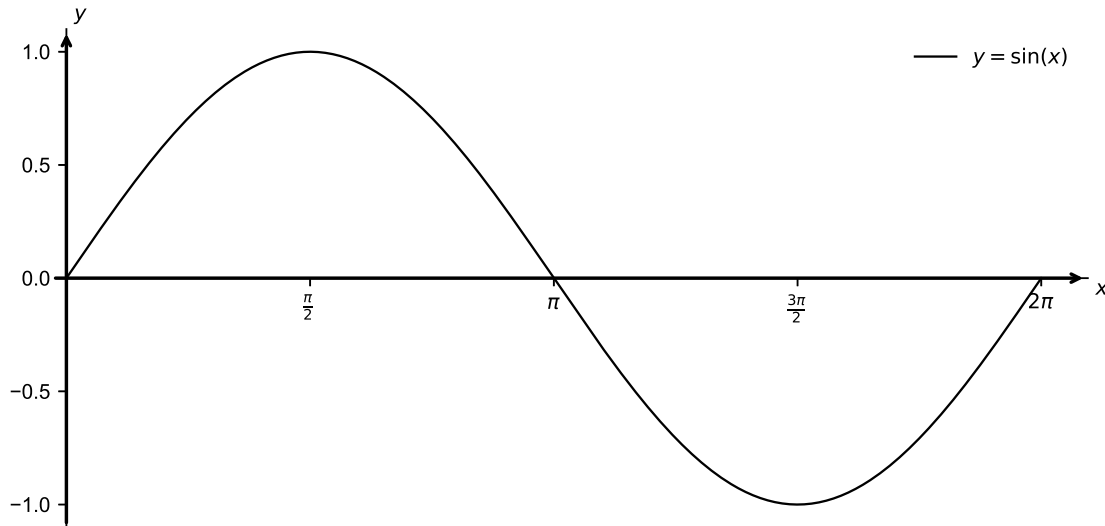


Fig. 3.1. Plot of the sine function. The Python program code of the plot is provided in Appendix A.

LaTeX automatically generates the list of figures after using the command `\listoffigures`. When using external materials (e.g., photos, plots, diagrams), the source must be indicated and care taken not to infringe copyright.

3.1.3. Tables

In an engineering thesis, it is useful to use tables (placed in the `table` environment) to present sets of technical parameters, measurement results, system configurations, or performance comparisons. The title is placed above the table (command `label`). If the table is extensive, it can be used the command `\resizebox{\textwidth}{!}{...}` or the `longtable` package for tables spanning multiple pages. Remember to refer to the table in the text using the command `\ref{}`.

Table. 3.1. Summary of the basic parameters of the test system

Parameter	Description	Value	Unit
Clock frequency	Main clock frequency of the microcontroller	80	MHz
Supply voltage	Value of the input voltage	5.0	V
Current consumption	Average current consumption during operation	120	mA
Operating temperature	Safe operating range of the device	−10 – 60	°C
Sampling time	Measurement interval in the main loop	50	ms
Number of input channels	Number of supported analogue sensors	8	–
Type of data transmission	Communication protocol used for transmission	UART	–

3.2. Chapter summary

The results chapter should end with a summary of the most important observations and conclusions drawn from the analysis. It is also worth noting the investigation's potential limitations and areas requiring further testing and comparison.

4. SUMMARY

This chapter should provide a brief summary of the entire thesis, highlighting the most important achievements, results, and conclusions from the research or project implementation. The purpose of this part is to present the extent to which the assumed objectives have been achieved and what practical or scientific significance the obtained results have.

The summary should also include a reflection on the difficulties and limitations encountered that may have affected the course of work or the results. It is also advisable to formulate suggestions regarding possible directions for further research, project development, or improvement.

4.1. Conclusions

The conclusions constitute a concise compilation of the key observations and results obtained in the thesis. They should be presented as an organised list, with an emphasis on the practical significance of the results and their consistency with the previously stated objectives and hypotheses.

4.2. The way forward

This subsection should indicate possible directions for expanding, optimising, or applying the developed solution. It may also describe:

- potential functional or technical improvements
- the possibility of integration with other systems
- application of the results in engineering, scientific, or industrial practice
- further studies or tests enabling verification of the effectiveness of the proposed approach

To conclude, this chapter should constitute the final, logical stage of the thesis, presenting its contribution and indicating how it can be further developed or used in the future.

BIBLIOGRAPHY

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APPENDIX A: GENERATING THE SINE FUNCTION

Appendices include supplementary materials that are not essential for understanding the main content of the thesis, but constitute a valuable extension. These may include, among others, fragments of source code, additional plots, summaries of measurement results, user manuals of the developed system, or auxiliary data used in the analysis.

A.1. Source Code

This appendix presents a sample source code that generates a plot of the *sine* function. The code can be executed in a Python environment (e.g., using the NumPy [4] and Matplotlib libraries). The example illustrates how to create simple plots of periodic functions to visualise results in an engineering thesis.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # === GENERAL PARAMETERS ===
5 fig_width = 6.7 # inches ~17cm (full A4 width)
6 fig_height = 3.2 # plot aspect
7 font_size = 9
8
9 plt.rcParams.update({
10     "font.family": "Arial",
11     "font.size": font_size,
12     "axes.spines.top": False,
13     "axes.spines.right": False,
14     "axes.linewidth": 0.8,
15     "mathtext.fontset": "dejavusans", # matches Arial
16 })
17
18 # === DATA ===
19 x = np.linspace(0, 2*np.pi, 400)
20 y = np.sin(x)
21
22 # === PLOT ===
23 fig, ax = plt.subplots(figsize=(fig_width, fig_height))
24 ax.plot(x, y, color='black', linewidth=1.0, label=r'$y = \sin(x)$')
25
26 # === AXIS SETTINGS ===
27 # Axes positioned at bottom and left
28 ax.spines['bottom'].set_position(('data', 0))
29 ax.spines['left'].set_position(('data', 0))
30 ax.spines['bottom'].set_color('black')
31 ax.spines['left'].set_color('black')
32 ax.spines['top'].set_visible(False)
33 ax.spines['right'].set_visible(False)
34
```



```

35 # === ARROWS ON AXES ===
36 arrow_style = dict(arrowstyle='->', color='black', lw=1.5)
37 ax.annotate('', xy=(2*np.pi + 0.3, 0), xytext=(-0.1, 0), arrowprops=arrow_style,
   ↪ clip_on=False)
38 ax.annotate('', xy=(0, 1.1), xytext=(0, -1.1), arrowprops=arrow_style,
   ↪ clip_on=False)
39
40 # === RANGE AND TICKS ===
41 ax.set_xlim(0, 2*np.pi + 0.3)
42 ax.set_ylim(-1.1, 1.1)
43 ax.set_xticks([np.pi/2, np.pi, 3*np.pi/2, 2*np.pi])
44 ax.set_xticklabels([r'\frac{\pi}{2}', r'\pi', r'\frac{3\pi}{2}',
   ↪ r'2\pi'])
45 ax.set_yticks([-1, -0.5, 0, 0.5, 1])
46
47 # === LABELS ===
48 ax.text(2*np.pi + 0.35, -0.05, r'$x$', fontsize=font_size, va='center')
49 ax.text(0.05, 1.12, r'$y$', fontsize=font_size, va='bottom')
50
51 # === FUNCTION LABEL ===
52 ax.legend(loc='upper right', frameon=False, fontsize=font_size)
53
54 # === APPEARANCE AND SAVE ===
55 ax.grid(False)
56 plt.tight_layout(pad=0.2)
57 plt.savefig("sinus.eps", format='eps', bbox_inches='tight')
58 plt.show()

```

Listing A.1. Generating a plot of the sine function in Python.

A.1.1. Notes on Code Formatting

Source code should be placed in the `lstlisting` environment, which preserves formatting and indentation and provides syntax highlighting for the language. If the code is extended, the inserted fragment may be limited, e.g.: `\lstinputlisting[firstline=10,lastline=60]{*.py}`. In engineering theses, it is recommended to include in the appendices only those fragments that are essential for understanding the solution; longer listings are best placed in an online repository, with a reference provided in the text (e.g., “The source code is available in the [GitHub repository](#)”).